

SUMMARY

Quantitative data scientist at an options trading firm. I backtest options strategies across NSE, BSE, MCX and US markets and built the Python / Polars / DuckDB pipeline behind the research — 1B+ rows, load times cut from hours to seconds

EXPERIENCE

Jr. Data Scientist — Pranjali Grow Cap Pvt Ltd ,Delhi.

Sep 2025 – Present

- Identified the firm's top-performing options setups — measured by backtested PnL across **days-to-expiry, weekday and strike** — by designing and testing **quantitative trading strategies** for NSE, BSE, MCX and US **derivatives** markets.
- Cut strategy research turnaround **from hours to seconds** on **100M–1B+ row** options datasets by replacing full Excel reloads with an **incremental Python + Polars pipeline** that re-processes only new market data.
- Enabled **near real-time strategy analysis** for the trading team — dashboards refresh in seconds instead of hours — by building a **DuckDB-over-Parquet** query layer and a **Streamlit** app with an Excel-style drag-and-drop pivot builder and **automated Excel** heatmap generation (xlwings).
- Optimized** strategy selection across **thousands of parameter combinations** per backtest — evaluating **profit potential and risk exposure** of each variant under diverse market conditions — via a systematic PnL aggregation framework that separates robust setups from noise.
- Automated daily data updates**, eliminating manual reloads entirely, by engineering memory-optimized ingestion (RAR → Parquet, chunked aggregation) that processes millions of rows per run within tight memory limits.

SKILLS

Quant & Markets:	Options & derivatives, backtesting, strategy optimization, PnL & risk analysis
Core:	Python, C++, SQL, statistics, probability
Data at scale:	Polars, DuckDB, Parquet, pandas, NumPy, Streamlit
Also:	scikit-learn, Power BI, Excel (xlwings), Git, Jupyter

SELECTED PROJECTS

- CAS Live Index — NIFTY/SENSEX reconstruction through the closing-auction blackout.** Built a live system that rebuilds both indices from constituent auction data (free-float weights; divisor recovery for SENSEX) using only public NSE/BSE endpoints, during the 15-minute window when official index feeds freeze. Tracked NSE's indicative close within ~10 points (0.04% MAE) across 10 live sessions, with per-stock attribution the exchange doesn't publish. [\(Link\)](#)
- Deep Learning for Options Trading — replication of Tan, Roberts & Zohren (2024).** Reimplemented the paper's end-to-end Sharpe-loss models (Linear/MLP/LSTM) on NIFTY/BANKNIFTY straddles; expanding walk-forward backtests against 8 classical baselines with transaction-cost sensitivity to 50 bps, and a documented audit of why headline Sharpe (3.37) should be discounted. . [\(Link\)](#)
- NLP Sentiment Risk Filter for Index Straddle Selling.** GDELT macro-news sentiment + FinBERT + walk-forward XGBoost classifying dangerous high-volatility days for NIFTY/BANKNIFTY option selling — 64% walk-forward accuracy vs ~57% base rate, with flagged skip-days showing larger next-day moves. [\(Link\)](#)

EDUCATION

Bachelor of Computer Applications : Guru Gobind Singh Indraprastha University · GPA 8/10

2022 – 2025

CERTIFICATION & ACHIEVEMENTS

Machine learning I (**Columbia+**), Deloitte Data Analytics Job Simulation (**Forage**), **1st place**, Glitch Verse coding competition (2025) · **Matrix 2.0 Hackathon**, IITM Janakpuri (2024)